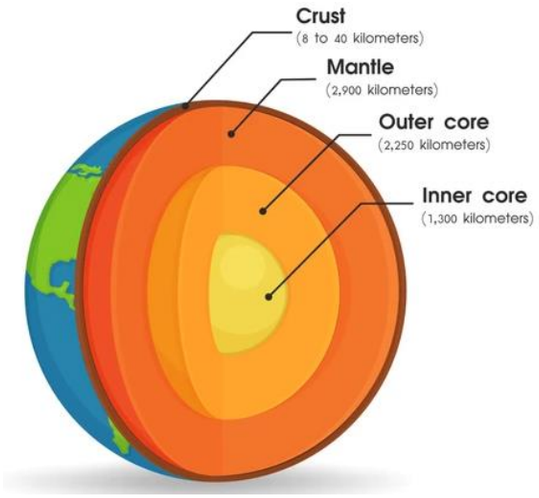
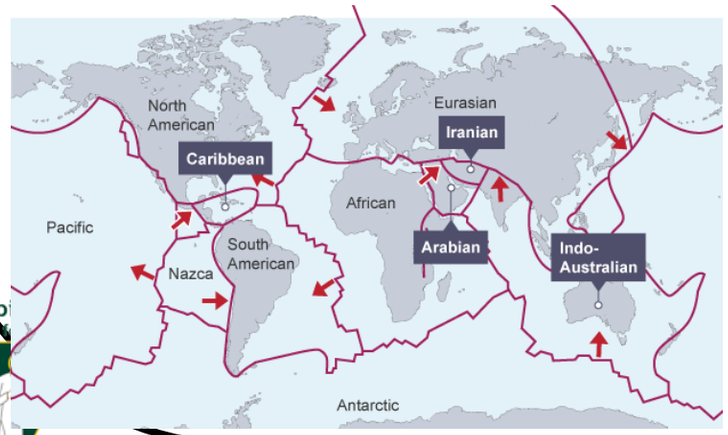
Knowledge Organiser
Year 8: Restless Earth – Structure of the Earth

What is the internal structure of the Earth?
- **Crust:** The thin, outermost layer; solid rock
- **Mantle:** Below the crust; semi-solid rock, it flows slowly
- **Outer Core:** Liquid layer composed of iron and nickel
- I**nner Core:** Solid, dense centre made of iron and nickel; extremely hot and under immense pressure

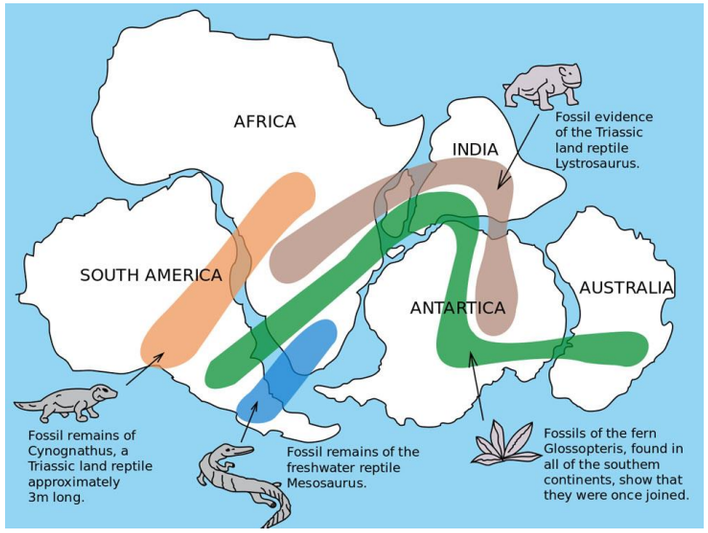


A diagram showing the approximate thickness of the layers of the Earth

What are tectonic plates?
- The crust is not a solid piece of rock, but is broken up into a number of **tectonic plates**
- Slowly moving semi-solid rocks in the mantle cause the tectonic plates to shift and move
- At the **boundaries** where different plates meet the plates can crash into each other or pull apart
- This can lead **to seismic processes** like volcanos and earthquakes



How do we know what is inside the Earth?
- The deepest hole ever dug by humans is 12km into the crust
- We cannot take samples from deep within the **mantle or core**
- But scientists can use waves to **predict** what the structure of the Earth is
- When a wave passes through a material it can **change speed or direction** depending on the **density** and **thickness** of the material
- By measuring the progress of **seismic waves** through the Earth's centre, we can **estimate** the size and makeup of each layer



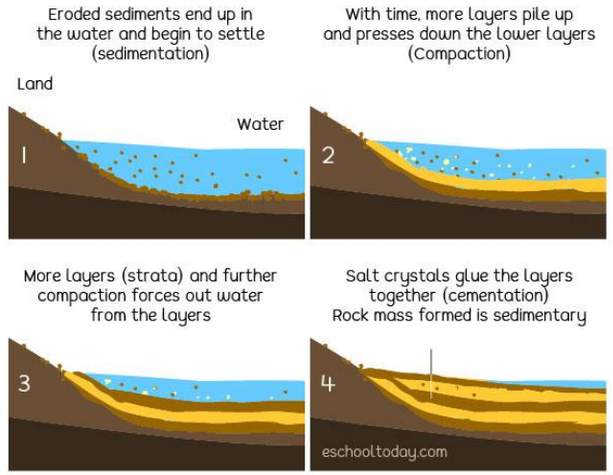
A diagram showing how fossils of the same species are found on different continents

Alfred Wegener and Continental Drift
- In 1912 the scientist **Alfred Wegener** proposed that continents could move
- He suggested that mountains formed when the edge of a drifting continent collided with another
- It wouldn't be till the 1960s that evidence of movement in the mantle was discovered and showed how this happens

Evidence for Wegener's theory:
1. The continents have shapes which "fit together" like jigsaw pieces
2. There are **similar rocks** on both sides of the Atlantic ocean
3. There are **similar fossils** on both sides of the Atlantic, including for animals that couldn't swim!

The three types of rocks:
- Rocks can be classified as either **sedimentary, igneous** or **metamorphic** depending on how they were formed
- Rocks can change from one form to another due to factors like **heat** and **pressure**

Rock	Examples	How it is formed
Sedimentary	Sandstone, limestone	Formed from the accumulation and compression of sediments over time.
Igneous	Basalt, obsidian	Formed from the cooling and solidification of magma or lava.
Metamorphic	Marble, slate	Formed from existing rocks that are changed by heat, pressure, or chemical processes.



A diagram showing the formation of sedimentary rocks from sediments under the ocean

RAG Yourself!

Learning Outcomes;	RAG
Describe the internal structure of the Earth	
Describe how tectonic plates behave	
Explain the evidence available for Wegener's theory of continental drift	
Explain how we know the structure of the Earth	
Name the three types of rocks	
Describe how sedimentary rocks are formed	

Knowledge Organiser
Year 8: Restless Earth – Structure of the Earth

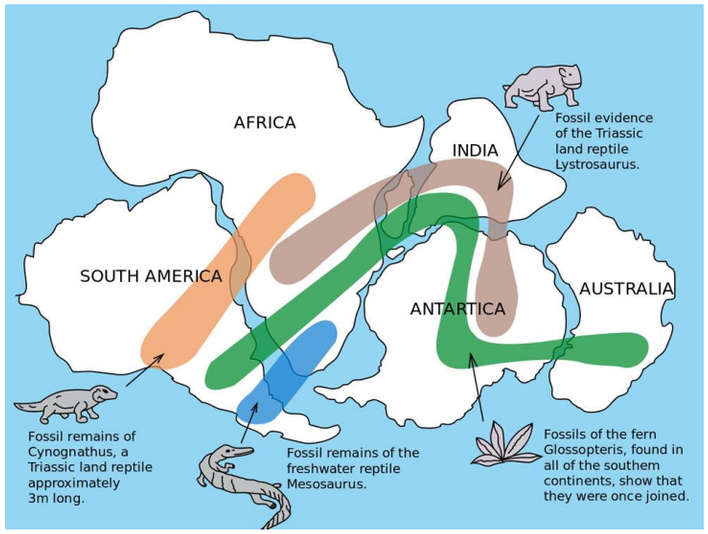
What is the internal structure of the Earth?

How do we know what is inside the Earth?

The three types of rocks:

Rock	Examples	How it is formed
Sedimentary		
Igneous		
Metamorphic		

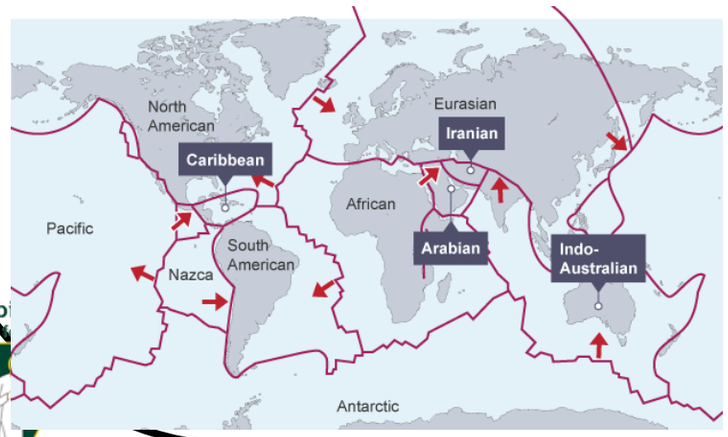
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A diagram showing the formation of sedimentary rocks from sediments under the ocean

What are tectonic plates?



Alfred Wegener and Continental Drift

Evidence for Wegener's theory:

RAG Yourself!

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